

Mohammad Mahmudul Hasan

Project Lead, AI Engineer

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Research Interests

	My research objective is to advance the field of 3D Urban Modeling by integrating Geospatial AI (GeoAI) with high-fidelity reconstruction techniques. I am specifically interested in investigating:
Core Focus	Autonomous Digital Twins: Leveraging Gaussian Splatting and CityGML to automate the generation of photorealistic, large-scale city models.
Methodology	Spatial Intelligence: Refining Point Cloud Analytics and Computer Vision pipelines for precise object localization and multi-temporal change detection in urban environments.
Vision	Developing scalable, Autonomous GIS Systems that bridge the gap between Remote Sensing and real-time spatial data automation.

Education

2017–2021	B.Sc. in Electrical and Electronic Engineering (EEE) , Islamic University of Technology (IUT), Bangladesh, <i>CGPA: 3.76/4.00 (First Class Honors)</i>
Thesis	<i>Quantifying Locomotive Features in EEG of Impaired Consciousness (Coma) with Distinctive Cerebral Rhythms.</i> Analyzed EEG time-series signals to identify discriminative features related to consciousness levels.
2014–2016	Higher Secondary Certificate (Science) , Notre Dame College, Dhaka, <i>GPA: 5.00/5.00</i> . Received Government Scholarship.
2012–2014	Secondary School Certificate (Science) , National Ideal College, Dhaka, <i>GPA: 5.00/5.00</i>

Professional Experience

2024–Present	Project Lead, AI Department , HawarIT Limited, Bangladesh. Leading AI and geospatial automation projects and managing multidisciplinary teams for GeoAI systems.
2023–2024	Senior AI Engineer , HawarIT Limited, Bangladesh. Led advanced AI system design for photogrammetry, point cloud analytics, and spatial automation.
2021–2023	ML Engineer , HawarIT Limited, Bangladesh. Developed machine learning and computer vision pipelines for geospatial applications.

Selected Technical & Research Projects

3D Intelligence	Developed AI-driven algorithms for semantic segmentation and BIM model extraction from large-scale aerial, mobile, and indoor LiDAR datasets. Implemented automated pipe detection and geometric property extraction, resulting in a 60% reduction in manual processing time with 95% extraction accuracy.
Localization	Engineered a comprehensive end-to-end GeoAI pipeline for the automated detection and geospatial localization of 130 traffic sign and 33 street furniture classes from 360° street-view imagery. Utilized YOLOv5 and custom tracking algorithms to achieve over 95% detection accuracy and 92% localization precision.
Change Detection	Developed a multi-temporal change detection system using deep learning to identify building extensions, new constructions, and demolitions from multi-year orthophotos. Integrated the system into a GIS-based web validation platform, achieving 88% accuracy and reducing manual inspection efforts by 65%.
City Modeling	Built an automated pipeline for large-scale 3D CityGML reconstruction and photorealistic texture generation from high-resolution aerial imagery. Designed custom multi-view texture projection algorithms considering occlusion and view selection, maintaining a texture shift of less than 3 pixels.
Orthophoto QA	Automated orthophoto production workflows by developing AI-based image quality assessment tools for cloud, shadow, and blur detection with 97% accuracy. Optimized seamline refinement and spatial validation using PostgreSQL/PostGIS and AHN point cloud data, cutting manual editing by 65%.

Pose Estimation	Developed a camera pose estimation framework utilizing feature matching and tie-point generation to align new aerial imagery with historical orthophotos. Achieved a 2-pixel alignment accuracy, which effectively reduced manual aerial triangulation efforts by 90%.
Remote Sensing	Deployed AI solutions using Sentinel multispectral imagery for multi-class crop classification, water quality assessment, and algae bloom detection. This innovative earth observation analytics framework secured 2nd place in an international innovation tender.
Signal Analysis	Engineered an automated machine fault detection method using MATLAB signal processing to analyze accelerometer data. Successfully identified and localized four distinct fault types and automated the reporting process.
Astrophysics	Processed and cleaned fragmented WMAP datasets to remove noise from Cosmic Microwave Background Radiation (CMBR). Developed algorithms to isolate primordial signals and produce clean sky-maps for astronomical analysis.

Publication

- 2021** Hasan, M.M., et al. "Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI." *2021 IEEE Asia-Pacific Conference on CSDE*. DOI: [10.1109/CSDE53843.2021.9718400](https://doi.org/10.1109/CSDE53843.2021.9718400).

Technical Skills

Programming	Python, SQL, JavaScript, C, MATLAB, R.
AI/Vision	PyTorch, TensorFlow, OpenCV, YOLO, FastAPI.
Geospatial	GDAL, GeoPandas, PostgreSQL/PostGIS, QGIS, ArcGIS.
3D/Point Cloud	Open3D, PCL, COLMAP, OpenMVS, CloudCompare.

Awards & Honors

- Excellent Performer (2024) and Borsho Shera (Best Performer of the Year (2022)), HawarIT Limited.
- 1st Place, MATLAB Competition, AUST.
- Top 15, Dhaka.AI Deep Learning Competition.
- 1st Prize, Nuclear Power Seminar, AUST.
- Government Scholarship, HSC.

References

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